



For 2026-2031, India has planned several high-value purchases, despite the indigenisation push

the SPECTRA electronic warfare suite, systems that collectively define the fighter's sensor fusion, survivability, and electronic combat architecture.

The entire modern Active Electronically Scanned Array radar and electronic warfare ecosystem has a common denominator: Gallium Nitride (GaN). The single material can bring many ambitious Indian defence programmes to a standstill. India had sought this GaN chip technology from France with the first tranche of 36 Rafale fighter jets it purchased. However, France refused to transfer it under the offset clause. The DRDO scientists cracked the GaN fabrication in 2023 and established its production in 2024.

India has also developed its own AESA radar named the Uttam. It was declared production-ready in 2023. But equipping Indian fighter jets is going to be an arduous task. The

first 40 LCA Mk1A fighter jets will be equipped with Israel EL/M-2052 radars; HAL intends to integrate the indigenous Uttam radars from the 41st unit onwards. The indigenous radar will feature in the second batch of 93 LCAMK1A.

India's electronics dependency has three layers—first is hardware (GaN chips fabrication), the second is system-level integration (putting indigenous radar into weapon platforms like LCA, Su-30MKIs and future warships) and third is software sovereignty. Out of these three, software sovereignty, as indicated by the no software access on imported platforms, seems to be the most difficult to achieve through indigenous efforts alone.

India is dependent on different countries for foundational defence manufacturing. Titanium, carbon fibre composites and special alloys are all imported.

Rajanikanth Balaraman, Co-Founder and Chief Growth and Technology Officer of Unimech Aerospace and Manufacturing, says: "These are highly controlled technologies with long development cycles, rigorous qualification standards and significant intellectual property barriers."

WHAT MUST BE DONE?

The Indian government has been signing defence contracts with technology transfer clauses. The Rafale deal delivered fighter jets but not deliver aero-engine independence. The future contracts should correct this.

Aero-engine development needs to be treated as a national moonshot. The Safran-GTRE partnership for the AMCA engine is encouraging. It must be funded and prioritised at the level India once accorded to its nuclear programme. India needs a dedicated Aero-Engine Mission, with ring-fenced funding and a single accountable programme director.

The India Semiconductor Mission and the defence acquisition process currently operate in separate silos. Every new defence platform contract should specify domestic chip content requirements on a phased timeline. This would create demand signals that pull semiconductor investment toward defence-grade specifications.

India should not just focus on the final assembly. It needs to build capability in titanium and superalloy processing, advanced composites, precision optical systems, and the speciality electronics that underpin every modern weapons system. This is unglamorous, long-cycle work that delivers no ribbon-cutting ceremony. It is also the work that determines whether the next generation of Indian defence platforms is genuinely independent of imports. **BT**